

SwarmOps Cloud

Test Plan and Test Report

Nima Sarayan

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Abstract

This document states how SwarmOps Cloud was tested and what the tests showed. The plan combines four levels: - automated Go tests that run against real MariaDB 11.4 and MySQL 8.4 servers; - HTTP integration tests; - static and rule tests of the web applications; - an end-to-end run in which a customer and an administrator used the system in a browser against a single-node Docker Swarm.

Every number reported here was produced by runs on 15 and 16 September 2026 and is quoted from their output. The report also lists the eleven defects the testing found — five of which were fixed during testing, each with its evidence — and states plainly what could not be verified in the local environment.

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1 Introduction

1.1 Purpose

The plan (sections 2–3) defines what is tested, how and when testing is complete. The report (sections 4–7) records the results. It is written for the course instructor and for anyone who must judge whether the system does what the Software Requirements Specification (SRS) requires.

1.2 Items under test

The items are the `university/cloud-platform` branch of SwarmOps, at the commits listed in Table 1, with the uncommitted documentation build.

Table 1: Commits under test

Commit	Content
d77aa59	Controller state in MySQL or MariaDB
40df03e	Commerce: wallet, orders, administrator review
ad37988	Storefront API and commerce administration routes
c4d0b26	Storefront and Sales area of the console
5cabba8	Fix: usage of a refunded deployment (defect D-2)
3e3d131	Fix: storefront changes after a session change (defect D-3)
be635c2	Fix: plan features, Persian plan text and storefront theme (defect D-11)

1.3 References

The Software Requirements Specification, the Software Architecture Document and the Database Design of this project; IEEE Std 829-2008 and ISO/IEC/IEEE 29119-3:2021 informed the structure.

2 Test plan

2.1 Approach

Table 2: Test levels

Level	What it establishes	Tools	Environment
Unit and store integration	Business rules and every store’s behaviour against a real database: transactions, constraints, locking, idempotency, migrations.	Go <code>testing</code> ; <code>sqltest</code> , which creates a throwaway schema per test	MariaDB 11.4 and MySQL 8.4 in containers

Level	What it establishes	Tools	Environment
HTTP integration	Routes, authentication, CSRF, authorisation, status codes and the storefront-to-console flow through the real handlers.	Go <code>httptest</code>	As above
Web	Types, console information-architecture rules, routing, formatting and builds.	TypeScript compiler, Node's test runner, Vite	Node 26
Operational checks	Stack definitions render, required documents exist, installer scripts parse, snapshot upgrade works.	<code>make</code> <code>stack-check-all</code> , <code>make docs-check</code> , <code>bash -n</code> , <code>migrate-state</code>	Local
End-to-end	The system as users experience it: a customer buys, an administrator confirms, the platform deploys, bills and supports.	Chromium, the running applications, direct SQL and Docker inspection	Single-node Swarm (section 3)

Tests of money and concurrency were written with the code they test and are designed to fail on the defects they guard against. D-2 below shows one such test failing before its fix and passing after.

2.2 Features to be tested

All functional requirements of the SRS (ACC, CAT, WAL, ORD, PRV, BIL, INV, SUP, REP, OPS, DAT) and the non-functional requirements for security, integrity, portability and usability. Section 5 traces each to its tests.

2.3 Features not to be tested

- Real payment processing (simulated by design).
- Certificate issuance from a public ACME authority.
- Multi-node Swarm scheduling.
- Performance and load. No throughput or latency targets are specified, and none are claimed.

2.4 Entry and exit criteria

Entry. The code builds and `go vet ./...` reports nothing.

Exit.

1. Every automated test passes on both database engines.
2. The web typecheck, tests and build succeed.

3. The operational checks succeed or are recorded as not runnable.
4. The end-to-end scenario completes, with every deviation recorded as a defect.
5. Every defect that affects money or security is fixed and re-tested.

2.5 Suspension and resumption

Testing stops when a defect makes further results meaningless — for example a ledger that no longer matches its balance. It resumes after the fix, from the first affected step.

3 Test environment

Table 3: Test environment

Component	Version or setting
Host	macOS laptop, Apple silicon (arm64)
Go	1.26
Node.js	26.5.0
Browser	Chromium in the development browser pane; Google Chrome 152 headless for screenshots
MariaDB	11.4, container on 127.0.0.1:3307
MySQL	8.4, container on 127.0.0.1:3308
Docker Engine	29.4.0 (OrbStack), single-node Swarm manager labelled <code>nim.edge=true</code>
Gateway Core	Traefik v3.6.13, installed through the console <code>go run ./cmd/api</code> with development authentication, mutations enabled, 127.0.0.1:8084
Machine agent	Source build stamped v0.22.0, loopback 127.0.0.1:9180
Web	Vite development server, 127.0.0.1:5284

The C4 Architecture Model (Figure 8) draws this environment.

4 Results of automated and operational tests

4.1 Summary

Table 4: Summary of results

Check	Result
<code>go vet ./...</code>	No findings (exit 0)
Go tests on MariaDB 11.4	33 packages pass, 3 packages have no tests; 570 top-level tests pass, 0 fail, 0 skipped
Go tests on MySQL 8.4	33 packages pass, 3 packages have no tests; 570 top-level tests pass, 0 fail, 0 skipped
Web typecheck (<code>tsc -b</code>)	0 errors
Web tests (<code>node --test src/*.test.mjs</code>)	69 pass, 0 fail
Web production build	Succeeds; the storefront and console are embedded in the binary
<code>make stack-check-all TAG=ci</code>	Every stack renders (exit 0)

Check	Result
<code>make docs-check</code>	Exit 0
Installer and deploy scripts, <code>bash -n</code>	16 scripts, 0 syntax errors
<code>shellcheck</code>	Not run: not installed on the test host
<code>migrate-state</code> on real snapshot files	Imported and re-imported without duplicates (section 4.4)

The counts are from the final run on 16 September 2026, after the fix for D-11; the run on 15 September, before it, passed 568 tests on each engine. Database-backed tests were run with `SWARMOPS_TEST_DATABASE_REQUIRED=1`, so a missing database fails the run instead of silently skipping tests.

4.2 Tests by package

Table 5: Top-level Go tests per package (identical on both engines)

Package	Tests	Package	Tests
<code>internal/ops</code>	106	<code>internal/cloud</code>	16
<code>api/http</code>	70	<code>internal/diagnosis</code>	14
<code>internal/queue</code>	45	<code>internal/k8simport</code>	13
<code>internal/agentcontrol</code>	43	<code>internal/nativectl</code>	13
<code>internal/agent</code>	39	<code>internal/warden</code>	13
<code>internal/source</code>	38	<code>internal/agentpull</code>	11
<code>internal/remote</code>	22	<code>internal/changepreview</code>	11
<code>scripts</code>	19	<code>internal/cli</code>	10
<code>cmd/swarmopsctl</code>	16	<code>internal/sqlstore</code>	9
<code>internal/config</code>	16	<code>internal/logstore</code>	6
		others (12 packages)	40

The web tests are in `operator-workflows.test.mjs` (27), `command-outcome.test.mjs` (22), `environment-editor.test.mjs` (10), `workspace-route.test.mjs` (4), `metric-series.test.mjs` (4) and `server-connection.test.mjs` (2).

4.3 Test cases for the new functionality

Table 6: Test cases for the new functionality

ID	Requirement	Test	Level	Result
TC-01	FR-ACC-1, FR-ACC-2, FR-ACC-4	<code>TestRegisterLoginAndSessionsAreSafe</code>	Integration	Pass
TC-02	FR-ACC-7	<code>TestASuspendedCustomerLosesEverySession</code>	Integration	Pass
TC-03	FR-WAL-1, NFR-REL-1	<code>TestConcurrentChargesNeverOverdrawAndMatchTheLedger</code> 50 parallel charges of 30,000 rials against 1,000,000 → exactly 33 succeed, 17 are refused, balance 10,000 equals the ledger	Integration	Pass
TC-04	FR-WAL-2	<code>TestATopUpIsCreditedOncePerIdempotencyKey</code>	Integration	Pass

ID	Requirement	Test	Level	Result
TC-05	FR-ORD-6	TestConfirmingAnOrderChargesProvisioningAndQueuesToGether	Integration	Pass
TC-06	FR-ORD-6	TestAConfirmationThatCannotQueueItsDeploymentChargesNothing	Integration	Pass
TC-07	FR-ORD-5	TestConfirmationRequiresTheReserve	Integration	Pass
TC-08	FR-ORD-7, FR-PRV-2	TestTheDeploymentOutcomeActivatesOrFailsAndRefundsOnce, extended for D-2: no usage and one correct invoice after a refund	Integration	Pass
TC-09	FR-BIL-1 to FR-BIL-3	TestBillingIsHourlyCappedAndIdempotent	Integration	Pass
TC-10	FR-BIL-4, FR-BIL-5	TestAWalletThatCannotPaySuspendsAndATopUpResumesWithoutBackCharging	Integration	Pass
TC-11	FR-INV-1 to FR-INV-4	TestInvoicesStateIncludedTaxAndAreIssuedOnce	Integration	Pass
TC-12	FR-ORD-1	TestOrdersRefuseTakenNamesAndInvalidApplications	Integration	Pass
TC-13	FR-SUP-3, NFR-SEC-2	TestTicketsAreVisibleOnlyToTheirCustomersAndAdministrators	Integration	Pass
TC-14	FR-REP-1, FR-REP-2	TestReportsReadTheViews	Integration	Pass
TC-15	FR-ACC-5, NFR-SEC-2, CI-1, CI-2	TestTheStorefrontSellsAndTheConsolePUTsHTTP. It checks: the public catalogue; a top-up refused without CSRF; top-up and order placement; 404 for another customer's order; a customer session refused on an operator route; an anonymous wallet refused; the pending queue; confirmation; the deployment appearing in the console's runs; the customer's projects and wallet after the first hour; and the confirmation being audited.	Integration	Pass
TC-16	FR-ORD-4	TestConfirmationRefusesAnUnknownServerHTTP	Integration	Pass
TC-17	FR-DAT-2	TestMigrationsAreContiguousAndSplitCIIntegrally, TestMigrateIsIdempotentAndRecordsEveryVersion, TestMigrateRefusesAModifiedAppliedMigration, TestMigrateRefusesADatabaseNewerThanTheController	Integration	Pass
TC-18	FR-DAT-4	TestSealedColumnIsBoundToTableColumnAndRow	Integration	Pass
TC-19	NFR-REL-2	TestWithTxRetriesADeadlockVictim: a real deadlock is forced between two transactions and the victim completes on retry	Integration	Pass
TC-20	NFR-PERF-2	TestOnlyOneCommandRunsAtATimeAcrossStores, TestAServerIsLeasedOneCommandAtATimeAcrossStores, TestClaimDueRollsBackWhenTheTransactionFails	Integration	Pass
TC-21	FR-DAT-3	TestImportFilesCopiesTheSealedLedgerWithItsTrailAndLog (queue), TestImportFilesCopiesSealedHistoryInOrderAndIsRepeatable (audit), and the import tests of every other store	Integration	Pass
TC-22	UI-1	TestStorefrontAddressesServeTheStorefrontPage	Web	Pass
TC-23	UI-2, C-5	operator-workflows.test.mjs: Sales area registered, every page routed, every API reading drawn	Web	Pass

ID	Requirement	Test	Level	Result
TC-24	FR-CAT-1, FR-CAT-2, UI-3	TestPlansKeepEnglishAndPersianText, TestSeededPlansOnlyPromiseWhatAnOrderProvides (D-11)	Integration	Pass

4.4 Upgrade from snapshot files

To test `migrate-state` on data the previous version wrote, not on fixtures, the release `v0.22.0` (commit `8ab10ed`, before SQL) was built from a separate worktree and run with its own data directory, connected to the machine agent. An administrator signed in. It wrote five real sealed snapshot files: core topology, agent registry, audit, servers and server keys.

`migrate-state` from the branch under test was then run twice against a copy of that directory and an empty database:

Table 7: Result of migrate-state on real snapshot files

Store	First run	Second run
Core topology members	1	1 (upserted, no duplicate)
Server profiles and keys	1 profile, 1 key	1 profile, 1 key (upserted)
Audit events	1	0 (already present)
Applications, credentials, source, routing, commands	0 (no files)	0

After both runs the database held exactly: one server (`Local machine`, `agent_api`, `Swarm active`), one server key, one audit event (`auth.login` by `admin`), one core member, one authority row and one agent certificate authority. The schema was at version 8. The command reported that the sealed files were unchanged, and they were still present. After migration 0009 was added, the test was repeated on 16 September with a fresh database and the same snapshot files: the same rows were imported, the second run again added none, and the schema reached version 9.

5 End-to-end test

5.1 Scenario

The end-to-end test followed the complete journey of the SRS use cases UC-01 to UC-10 with a test customer, *Sara Ahmadi* (`sara.demo@example.com`, a fictitious address), and the administrator `admin`. The HTTP statuses below are those recorded by the browser. Database and Docker observations were read directly. Times are UTC on 15 September 2026.

5.2 Execution log

Table 8: End-to-end execution log

Step	Time	Action	Observed result	Verdict
E-01	16:43	Customer registers and is signed in.	POST <code>/auth/register</code> 201, POST <code>/auth/login</code> 200; wallet page shows 0 toman.	Pass

Step	Time	Action	Observed result	Verdict
E-02	16:43	Customer tops up 500,000 toman.	POST <code>/wallet/topups</code> 200; balance 500,000 toman; ledger row <code>topup</code> +5,000,000 rials.	Pass
E-03	16:45	Customer orders <i>Starter</i> for sara-shop (image <code>nginxinc/nginx-unprivileged:1.27-alpine</code> , port 8080).	POST <code>/orders</code> 201; order 1 <i>In review</i> .	Pass
E-04	16:50	Administrator opens Sales › Orders › Review with no Swarm manager available.	No server offered; confirmation impossible (FR-ORD-4). Swarm was then initialised on the test host, with the owner’s approval.	Pass
E-05	16:56	Administrator confirms order 1 on <i>Local machine</i> with <code>CONFIRM_ORDER_1</code> .	POST <code>/orders/1/confirm</code> 200; ledger <code>order_charge</code> −4,000; deployment command queued. Deployment failed three times with <code>gateway_required</code> , since no gateway was installed; command <code>needs_attention</code> ; order and project <code>failed</code> ; <code>refund</code> +4,000; balance restored to 5,000,000 rials.	Pass (failure path)
E-06	17:00–17:05	Administrator prepares and installs the gateway: sets the ACME email and dashboard host, runs <i>Fix all 4 prerequisites</i> , then <i>Install gateway</i> with <code>DEPLOY_TRAEFIK</code> .	Repair command <code>succeeded</code> (overlay network <code>traefik</code> , label <code>nim.edge=true</code> , dynamic config, dashboard secret created); install <code>succeeded</code> ; <code>traefik_traefik</code> 1/1. Deviations D-6 and D-7.	Pass with deviations
E-07	17:06	Customer orders sara-blog with the same nginx image; administrator confirms.	Confirm 200; charge −4,000; stack deployed; task never healthy, because GET <code>/healthz</code> returned 404. After the 10-minute execution timeout the command became <code>needs_attention</code> , order <code>failed</code> , <code>refund</code> +4,000 at 17:16:48. The queue was blocked meanwhile (D-5). Checkout did not state the health requirement (D-1).	Pass (failure path) with deviations
E-08	17:11–17:18	Customer orders sara-api with <code>mendhak/http-https-echo:41</code> ; administrator confirms.	Confirm 200 at 17:12; charge −4,000; command ran once order 2 released the <code>production-sara-api_app</code> 1/1, health <i>healthy</i> ; command <code>succeeded</code> ; order and project <code>active</code> .	Pass
E-09	17:20	Customer views Projects.	sara-api <i>Active</i> ; the failed projects still showed 400 toman <i>charged this month</i> , although refunded (D-2); the card’s date label was unclear (D-4).	Fail → fixed

Step	Time	Action	Observed result	Verdict
E-10	18:02	Hourly billing, with no user action.	Usage row for sara-api , hour 18:00, 4,000 rials, created at 18:02:15 by the five-minute billing job.	Pass
E-11	17:26	Customer opens a support ticket after the core was restarted.	<code>POST /tickets 403 Invalid request token</code> (D-3).	Fail → fixed
E-12	17:33	Re-test of E-11 after the fix, with the page's token made stale by signing in again from another context.	<code>POST /tickets 403</code> → <code>GET /auth/me 200</code> → <code>POST /tickets 201</code> ; exactly one ticket row.	Pass
E-13	17:35	Administrator replies to the ticket in Sales › Support; customer reads it.	<code>POST /commerce/tickets/1/replies 200</code> ; ticket answered ; the reply is visible in the storefront.	Pass
E-14	18:05	Administrator issues invoices for 2026-09.	<code>POST /invoices/issue 200</code> , <code>{"issued": 1}</code> ; INV-202609-000001, total 16,000 rials, tax 1,454 rials (145 toman) at 10 %. The total includes the two refunded first hours recorded before the D-2 fix.	Pass — evidence of D-2
E-15	17:52	Ledger consistency and revenue report.	Balance equals ledger sum for every wallet (0 mismatches); <code>v_revenue_by_month</code> for 2026-09: charged 12,000, refunded 8,000, topped up 5,000,000, 1 paying customer.	Pass
E-16	18:13	Pages captured in English and Persian, at 1280 and 390 pixels, in light and dark themes.	23 screenshots. Review found that the seeded plans listed features an order does not provide, that the Persian storefront showed them in English, and that the storefront ignored the dark theme (D-11).	Fail → fixed
E-17	16 Sep, 20:47	Core restarted after the test host itself had been restarted.	Core re-pinned the rebuilt agent and connected. Its first billing run charged sara-api for the hours 19:00 and 20:00, during part of which the host was off (D-10).	Pass with deviation
E-18	16 Sep, 21:00	Plan pages, checkout, dashboard (dark) and console plans re-captured after the D-11 fix.	Accurate features in English and Persian; the storefront follows the dark theme.	Pass

The ledger at the end of 15 September, which reconciles exactly with the balance of 4,992,000 rials, was:

Table 9: Wallet ledger after the run

Kind	Amount (rials)	Balance after
topup	+5,000,000	5,000,000
order_charge (order 1)	−4,000	4,996,000
refund (order 1)	+4,000	5,000,000
order_charge (order 2)	−4,000	4,996,000
order_charge (order 3)	−4,000	4,992,000
refund (order 2)	+4,000	4,996,000
usage (order 3, 18:00)	−4,000	4,992,000

6 Defects

Table 10: Defects found during testing

ID	Severity	Found in	Description	Resolution
D-1	Medium	E-07	Checkout did not tell customers that the image must answer <code>GET /healthz</code> on its port and contain <code>wget</code> or <code>curl</code> , so a common image failed after being charged.	Fixed (c4d0b26): the hint states the requirement in English and Persian.
D-2	High	E-09, E-14	A failed deployment refunded its first hour but kept that hour’s usage row. The project kept showing the money as spent, and the month’s invoice billed hours already refunded.	Fixed (5cabba8): the usage row is deleted in the refund transaction. TC-08 was extended; with the fix reverted it fails (<i>a refunded project still shows 10000 rial of usage</i>), and with the fix it passes on both engines. Rows recorded before the fix remain in the test database, as E-14 shows.
D-3	High	E-11	After the session cookie changed (another sign-in), the storefront kept its old CSRF token, and every change failed with 403 until the page was reloaded.	Fixed (3e3d131): after a 403 the client reads the current session once and repeats the change with the same idempotency key. Verified in E-12.
D-4	Low	E-09	The project card labelled its date only as <i>Date</i> .	Fixed (5cabba8): <i>Running since</i> .
D-5	Medium	E-07	The worker runs one command at a time, and a deployment waits for its service to converge. An image that never becomes healthy blocked all other commands, including order 3’s deployment, until the ten-minute timeout.	Open , recorded in the Software Architecture Document, section 11.

ID	Severity	Found in	Description	Resolution
D-6	Low	E-06	The development agent reports version <code>dev</code> , which the gateway repair refuses (it requires v0.9.3 or later). A development-only condition: the agent was rebuilt with the release version stamped, as release builds are.	Not a product defect ; documented in the Installation Guide.
D-7	Low	E-06	Saving gateway static settings before the prerequisites exist queues a command that fails with <code>traefik_network_required</code> after eight attempts.	Open ; the console could require prerequisites first.
D-8	Low	review	A failed project keeps its application name, so the customer cannot reorder under that name.	Open , recorded as technical debt.
D-9	Low	E-13	Addresses of the form <code>#/support/<id></code> open the ticket list rather than the ticket.	Open , recorded as technical debt.
D-10	Medium	E-17	Billing charges a project for every elapsed hour while its status is <i>active</i> ; it does not check that the service was running. Hours during which the whole host was down were charged.	Open . A remedy is to charge only hours in which Swarm reports a running task.
D-11	High	E-16	The plans seeded by migration 0008 promised an HTTPS domain, metrics, tracing, managed databases and priority support, which a storefront order does not provide; the Persian storefront showed plan features in English; and the storefront did not follow the viewer’s dark theme.	Fixed (be635c2): migration 0009 replaces the unedited seeded features with accurate ones in English and Persian and adds Persian plan text; the storefront follows the system theme. Covered by TC-24 on both engines.

7 Coverage

Statement coverage was measured on MariaDB with `go test -cover` for the packages central to this project. Coverage shows which statements the tests executed, not whether their results were checked. The test cases above, which assert outcomes, are the primary evidence.

Table 11: Statement coverage on MariaDB 11.4

Package	Statement coverage
<code>internal/queue</code>	76.6 %
<code>internal/sqlstore</code>	75.0 %
<code>internal/audit</code>	74.6 %
<code>internal/cloud</code>	72.6 %
<code>api/http</code>	33.8 %

`api/http` is the lowest because it holds all of SwarmOps' operations handlers, most of which predate this project and are tested at the level of the operations they call. The storefront and commerce routes are exercised end to end by TC-15 and TC-16.

8 What was not verified

The following were outside what the local environment could demonstrate. They are not claimed:

- **Public certificates.** No certificate was issued: the gateway used a non-public host name. The ACME configuration was stored and rendered.
- **Multi-node Swarm.** All tests ran on a single node, so placement across workers, node failure and manager quorum were not exercised.
- **The production installation.** The systemd service units, protected secret files and the Swarm stack for Core were checked only by rendering (`stack-check-all`) and by `bash -n`. They were not installed on an Ubuntu host.
- **End-to-end on MySQL.** The browser run used MariaDB. MySQL 8.4 was covered by the full automated suite.
- **Continuous integration.** The workflow now runs the suite against MariaDB and MySQL, but no hosted CI run was observed, because the branch was not pushed.
- **Performance.** No load or latency measurements were made.
- **Payments.** Top-ups are simulated by design.
- **shellcheck.** It was not available on the host.

9 Conclusion

All exit criteria but one were met. Every automated test passed on both database engines (570 each), and the web checks, operational checks and upgrade test succeeded. The end-to-end scenario covered each principal use case, including both failure paths of provisioning. The three high-severity defects testing found, D-2, D-3 and D-11, were fixed and re-tested. The one criterion not fully met is the operational checks: `shellcheck` could not be run and is recorded as such. The open defects are of low or medium severity and are recorded with remedies.